

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks    |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a)(i)  | <p>1 mark for security method, 2 marks for description</p> <p><b>Anti-virus (software) // Anti-malware (software)</b></p> <ul style="list-style-type: none"> <li>Scans the computer system (for viruses)</li> <li>Has a record of known viruses</li> <li>Removes/quarantines any viruses that are found</li> <li>Checks data before it is downloaded</li> <li>... and stops download if virus found/warns user may contain virus</li> </ul> <p><b>Firewall // Proxy server</b></p> <ul style="list-style-type: none"> <li>Monitors traffic coming <b>into and out of</b> the computer system</li> <li><b>Checks</b> that the traffic meets any criteria/rules set</li> <li>Blocks any traffic that does not meet the criteria/rules set // set blacklist/whitelist</li> </ul> | <b>3</b> |

**PUBLISHED**

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a)(ii) | <p>1 mark for security method, 2 marks for description</p> <p><b>Firewall // proxy server</b></p> <ul style="list-style-type: none"> <li>Monitors traffic coming into and out of the computer system</li> <li>Check that the traffic meets any criteria/rules set</li> <li>Blocks any traffic that does not meet the criteria/rules set // set blacklist/whitelist</li> </ul> <p><b>NOTE: Cannot be awarded if already given in 4(a)(i)</b></p> <p><b>Passwords</b></p> <ul style="list-style-type: none"> <li>Making a password stronger // by example</li> <li>Changing it regularly</li> <li>Lock out after set number of attempts // stops brute force attacks // makes it more difficult to guess</li> </ul> <p><b>Biometrics</b></p> <ul style="list-style-type: none"> <li>Data needed to enter is unique to individual</li> <li>... therefore very difficult to replicate</li> <li>Lock out after set number of attempts</li> </ul> <p><b>Two-step verification // Two-factor authentication</b></p> <ul style="list-style-type: none"> <li>Extra data is sent to device, pre-set by user</li> <li>... making it more difficult for hacker to obtain it</li> <li>Data has to be entered into the same system</li> <li>... so if attempted from a remote location, it will not be accepted</li> </ul> | <b>3</b> |

**REMOVED**

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks    |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a)(iii) | <p>1 mark for security method, 2 marks for description</p> <p><b>Anti-spyware software // Anti-malware (software)</b></p> <ul style="list-style-type: none"> <li>Scans the computer for spyware</li> <li>Removes/quarantines any spyware that is found</li> <li>Can prevent spyware being downloaded</li> </ul> <p><b>NOTE: Anti-malware (software) cannot be awarded if already given in 4(a)(i)</b></p> <p><b>Drop-down boxes // onscreen/virtual keyboard</b></p> <ul style="list-style-type: none"> <li>Means key logger cannot collect data // key presses cannot be recorded</li> <li>... and relay it to third party</li> </ul> <p><b>Two-step verification // Two-factor authentication</b></p> <ul style="list-style-type: none"> <li>Extra data is sent to device, pre-set by user</li> <li>... making it more difficult for hacker to obtain it</li> <li>Data has to be entered into the same system</li> <li>... so if attempted from a remote location, it will not be accepted</li> </ul> <p><b>NOTE: Cannot be awarded if already given in 4(a)(ii)</b></p> <p><b>Firewall // proxy server</b></p> <ul style="list-style-type: none"> <li>Monitors traffic coming into and out of the computer system</li> <li>Check that the traffic meets any criteria/rules set</li> <li>Blocks any traffic that does not meet the criteria/rules set // set blacklist/whitelist</li> </ul> <p><b>NOTE: Cannot be awarded if already given in 4(a)(i) or 4(a)(ii)</b></p> | <b>3</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks    |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6(a)     | <p><b>Four</b> from (max 2 marks per improvement):</p> <ul style="list-style-type: none"> <li>• Make the password require more characters</li> <li>• Makes the password harder to crack/guess</li> <li>• More possible combinations for the password</li> </ul><br><ul style="list-style-type: none"> <li>• Make the password require different types of characters</li> <li>• Makes the password harder to crack/guess</li> <li>• More possible combinations for the password</li> </ul><br><ul style="list-style-type: none"> <li>• Use a biometric device</li> <li>• Hard to fake a person's biological data // data is <b>unique</b></li> </ul><br><ul style="list-style-type: none"> <li>• Two-step verification // Two factor-authentication</li> <li>• Adds an additional level to hack</li> <li>• Have to have the set device for the code to receive it</li> <li>• Drop-down boxes // onscreen keyboard</li> <li>• To prevent passwords being obtained using keylogger</li> </ul><br><ul style="list-style-type: none"> <li>• Request random characters</li> <li>• Won't reveal entire password</li> </ul><br><ul style="list-style-type: none"> <li>• Set number of password attempts</li> <li>• Will lock account if attempting to guess</li> <li>• Will stop brute-force attacks</li> </ul> | <b>4</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 8(a)     | <p><b>Six</b> from:</p> <ul style="list-style-type: none"> <li>• SSL is a (security) protocol</li> <li>• It encrypts any data that is sent</li> <li>• It uses/sends digital certificates ...</li> <li>• ... which is sent to the (buyer's/user's) browser // requested by (buyer's/user's) browser</li> <li>• ... that contains the gallery's public key</li> <li>• ... that can be used to authenticate the gallery</li> <li>• Once the certificate is authenticated, the transaction will begin</li> </ul> | <b>6</b> |

| Question                                                        | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks        |             |              |                                                  |  |   |                                                         |   |  |                                                         |  |   |                                                   |  |   |                                                                 |  |   |                                                           |   |  |          |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|--------------|--------------------------------------------------|--|---|---------------------------------------------------------|---|--|---------------------------------------------------------|--|---|---------------------------------------------------|--|---|-----------------------------------------------------------------|--|---|-----------------------------------------------------------|---|--|----------|
| 8(b)                                                            | <p>1 mark for each correct tick.</p> <table border="1"> <thead> <tr> <th>Statement</th><th>True<br/>(✓)</th><th>False<br/>(✓)</th></tr> </thead> <tbody> <tr> <td>Firewalls are only available as hardware devices</td><td></td><td>✓</td></tr> <tr> <td>Firewalls allow a user to set rules for network traffic</td><td>✓</td><td></td></tr> <tr> <td>Firewalls will automatically stop all malicious traffic</td><td></td><td>✓</td></tr> <tr> <td>Firewalls only examine traffic entering a network</td><td></td><td>✓</td></tr> <tr> <td>Firewalls encrypt all data that is transmitted around a network</td><td></td><td>✓</td></tr> <tr> <td>Firewalls can be used to block access to certain websites</td><td>✓</td><td></td></tr> </tbody> </table> | Statement    | True<br>(✓) | False<br>(✓) | Firewalls are only available as hardware devices |  | ✓ | Firewalls allow a user to set rules for network traffic | ✓ |  | Firewalls will automatically stop all malicious traffic |  | ✓ | Firewalls only examine traffic entering a network |  | ✓ | Firewalls encrypt all data that is transmitted around a network |  | ✓ | Firewalls can be used to block access to certain websites | ✓ |  | <b>6</b> |
| Statement                                                       | True<br>(✓)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | False<br>(✓) |             |              |                                                  |  |   |                                                         |   |  |                                                         |  |   |                                                   |  |   |                                                                 |  |   |                                                           |   |  |          |
| Firewalls are only available as hardware devices                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ✓            |             |              |                                                  |  |   |                                                         |   |  |                                                         |  |   |                                                   |  |   |                                                                 |  |   |                                                           |   |  |          |
| Firewalls allow a user to set rules for network traffic         | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |             |              |                                                  |  |   |                                                         |   |  |                                                         |  |   |                                                   |  |   |                                                                 |  |   |                                                           |   |  |          |
| Firewalls will automatically stop all malicious traffic         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ✓            |             |              |                                                  |  |   |                                                         |   |  |                                                         |  |   |                                                   |  |   |                                                                 |  |   |                                                           |   |  |          |
| Firewalls only examine traffic entering a network               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ✓            |             |              |                                                  |  |   |                                                         |   |  |                                                         |  |   |                                                   |  |   |                                                                 |  |   |                                                           |   |  |          |
| Firewalls encrypt all data that is transmitted around a network |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ✓            |             |              |                                                  |  |   |                                                         |   |  |                                                         |  |   |                                                   |  |   |                                                                 |  |   |                                                           |   |  |          |
| Firewalls can be used to block access to certain websites       | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |             |              |                                                  |  |   |                                                         |   |  |                                                         |  |   |                                                   |  |   |                                                                 |  |   |                                                           |   |  |          |

# 0478/12 – JUNE 2019

| Question                                                                  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks        |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|--------------|---------------------------------------------------------------|---|--|---------------------------------------------------------------------------|---|--|--------------------------------------------------------|--|---|---------------------------------------------------|---|--|---------------------------------------------------|---|--|---|
| 5                                                                         | <ul style="list-style-type: none"> <li>– Password protection</li> <li>– <b>Password</b> is released on the <b>release date</b></li> <li>– Encryption</li> <li>– Encryption <b>key</b> is released on the <b>release date</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                       | 4            |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| 6(b)                                                                      | <p>1 mark per each correct row.</p> <table> <tr> <th>Statement</th><th>True<br/>(✓)</th><th>False<br/>(✓)</th></tr> <tr> <td>Cookies can be used to store a customer's credit card details</td><td>✓</td><td></td></tr> <tr> <td>Cookies can be used to track the items a customer has viewed on a website</td><td>✓</td><td></td></tr> <tr> <td>Cookies will corrupt the data on a customer's computer</td><td></td><td>✓</td></tr> <tr> <td>Cookies are downloaded onto a customer's computer</td><td>✓</td><td></td></tr> <tr> <td>Cookies can be deleted from a customer's computer</td><td>✓</td><td></td></tr> </table> | Statement    | True<br>(✓) | False<br>(✓) | Cookies can be used to store a customer's credit card details | ✓ |  | Cookies can be used to track the items a customer has viewed on a website | ✓ |  | Cookies will corrupt the data on a customer's computer |  | ✓ | Cookies are downloaded onto a customer's computer | ✓ |  | Cookies can be deleted from a customer's computer | ✓ |  | 5 |
| Statement                                                                 | True<br>(✓)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | False<br>(✓) |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| Cookies can be used to store a customer's credit card details             | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| Cookies can be used to track the items a customer has viewed on a website | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| Cookies will corrupt the data on a customer's computer                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ✓            |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| Cookies are downloaded onto a customer's computer                         | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| Cookies can be deleted from a customer's computer                         | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |              |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| 6(d)                                                                      | <p><b>One</b> from:</p> <ul style="list-style-type: none"> <li>– Uniform resource locator</li> <li>– The website's address</li> <li>– User friendly version of the IP address</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1            |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| 6(e)(i)                                                                   | <p><b>Four</b> from:</p> <ul style="list-style-type: none"> <li>– Designed to deny people access to a website</li> <li>– A large number/numerous requests are sent (to a server) ...</li> <li>– ... all at the same time</li> <li>– The server is unable to respond/struggles to respond to all the requests</li> <li>– The server fails/times out as a result</li> </ul>                                                                                                                                                                                                                                                     | 4            |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |
| 6(e)(ii)                                                                  | <p><b>One</b> from:</p> <ul style="list-style-type: none"> <li>– Proxy server</li> <li>– Firewall</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1            |             |              |                                                               |   |  |                                                                           |   |  |                                                        |  |   |                                                   |   |  |                                                   |   |  |   |

# 0478/13 – JUNE 2019

| PUBLISHED |                                                                                                                                                                                                                                                                                                                                                                                     |       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                              | Marks |
| 1(d)      | <p>1 mark for each correct term, in the correct place:</p> <ul style="list-style-type: none"> <li>– URL</li> <li>– https</li> <li>– Domain name</li> <li>– Web server</li> <li>– Browser</li> <li>– HTML</li> </ul>                                                                                                                                                                 | 6     |
| 1(e)(i)   | <ul style="list-style-type: none"> <li>– Small packets of data</li> <li>– ... that are stored by the web browser</li> </ul>                                                                                                                                                                                                                                                         | 2     |
| 1(e)(ii)  | <p><b>Four</b> from:</p> <ul style="list-style-type: none"> <li>– To store a customer's password ...</li> <li>– To store a customer's credit card details ...</li> <li>– ... so they do not need to be re-entered in future</li> <li>– To track what the customer has viewed on the website ...</li> <li>– ... so she can send them adverts that match their preferences</li> </ul> | 4     |

**PUBLISHED**

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(a)     | <b>Four from:</b> <ul style="list-style-type: none"> <li>- The company could use the firewall to set criteria</li> <li>- Gaming websites can be listed as blocked websites // ports can be blocked</li> <li>- The firewall would examine any traffic leaving the network</li> <li>- If it detected traffic requesting a listed website, it will block access to it</li> <li>- Keeps a log of all attempts to access blocked websites</li> </ul>                                                                                                                                                                                          | <b>4</b> |
| 3(c)     | <b>Six from:</b> <ul style="list-style-type: none"> <li>- The user could have been sent an email with an attachment / link containing the spyware</li> <li>- The user could have clicked a link on an untrusted website</li> <li>- When the attachment / link was clicked the spyware was downloaded onto the user's computer</li> <li>- The spyware recorded all the key logs from the user's keyboard</li> <li>- The recorded key logs were sent back to the creator of the spyware</li> <li>- The key logs were analysed</li> <li>- A common pattern / word in the key logs could have allowed a password to be identified</li> </ul> | <b>6</b> |



| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(a)     | <b>Three from:</b> <ul style="list-style-type: none"> <li>∞ Malicious software // type of malware</li> <li>∞ Tracks / records keypresses // keylogger</li> <li>∞ Sends data to third party</li> <li>∞ Collected data is analysed to obtain data</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>3</b> |
| 3(b)     | <b>One mark for identified method, one mark for how it prevents spyware:</b> <p><b>Drop-down boxes // onscreen / virtual keyboard</b></p> <ul style="list-style-type: none"> <li>∞ Means key logger cannot collect data</li> </ul> <p><b>Only requires part of the password</b></p> <ul style="list-style-type: none"> <li>∞ Hacker doesn't get the full password</li> </ul> <p><b>Two-step verification // Two-factor authentication</b></p> <ul style="list-style-type: none"> <li>∞ Extra data is sent to device making it more difficult for hacker to obtain it</li> <li>∞ Data has to be entered into the same system // if attempted from a remote location, it will not be accepted</li> </ul> <p><b>Use a biometric device</b></p> <ul style="list-style-type: none"> <li>∞ The person's biological data (e.g. their fingerprint) is also required</li> </ul> | <b>2</b> |
| 3(c)     | <b>Four from:</b> <ul style="list-style-type: none"> <li>∞ Monitors traffic coming into and out of the computer system</li> <li>∞ Checks that the traffic meets any criteria / rules set</li> <li>∞ Blocks any traffic that does not meet the criteria / rules set</li> <li>∞ Allows a set blacklist / whitelist // can block certain IP addresses</li> <li>∞ Can close certain ports</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>4</b> |
| 3(d)     | <b>Two from:</b> <ul style="list-style-type: none"> <li>∞ Passwords // biometrics</li> <li>∞ Levels of access</li> <li>∞ Proxy servers</li> <li>∞ Physical security methods – e.g. PC's in locked rooms, etc.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>2</b> |
| 6(b)(i)  | <b>One from:</b> <ul style="list-style-type: none"> <li>∞ Protocol is HTTPS</li> <li>∞ Padlock icon is locked</li> <li>∞ Can view website certificate</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>1</b> |
| 6(b)(ii) | <b>Five from:</b> <ul style="list-style-type: none"> <li>∞ Browser / client sends request to webserver to request identification</li> <li>∞ Web server sends its digital / security certificate</li> <li>∞ Browser authenticates certificate ...</li> <li>∞ ... if authentic connection, is established</li> <li>∞ Any data sent is encrypted ...</li> <li>∞ ... using public and private keys</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>5</b> |
| 9(b)     | <b>One mark for each correct term in the correct order:</b> <ul style="list-style-type: none"> <li>∞ browser</li> <li>∞ IP address</li> <li>∞ web server</li> <li>∞ HTML</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>4</b> |

# 0478/12 – NOVEMBER 2019

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| 7(b) | <b>Three from:</b> <ul style="list-style-type: none"> <li>∞ Display a web page</li> <li>∞ Sends a request to the web server</li> <li>∞ Receives data from web server</li> <li>∞ Translates HTML files</li> <li>∞ Processes client-side script, e.g. JavaScript</li> <li>∞ Store favourites</li> <li>∞ Store history</li> <li>∞ Navigation forward and backward</li> <li>∞ Check security</li> <li>∞ Store / access cookies</li> <li>∞ Find specific text within a web page</li> <li>∞ Downloading file from the web</li> <li>∞ Allows a homepage</li> <li>∞ Allows multiple tabs / web pages to be opened</li> <li>∞ Stores data in its cache</li> </ul> | 3 |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

| Question | Answer                                                                                                                                                                                                                                                                                                                    | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(c)     | <b>Three from:</b> <ul style="list-style-type: none"> <li>∞ Hypertext Transfer Protocol Secure // It is a protocol ...</li> <li>∞ ... that is a set of rules/standards</li> <li>∞ Secure version of <u>HTTP</u></li> <li>∞ Secure website // secures data</li> <li>∞ Uses TLS / SSL</li> <li>∞ Uses encryption</li> </ul> | 3     |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 10(b) | <b>Six from (maximum three marks per security method):</b> <ul style="list-style-type: none"> <li>∞ Firewall ...</li> <li>∞ ... Monitors the traffic</li> <li>∞ ... <b>Blocks</b> any traffic that doesn't meet the <b>criteria / rules</b></li> <li>∞ (Strong) password // biometric ...</li> <li>∞ ... Data cannot be accessed without the use of the password / bio data</li> <li>∞ ... Prevent brute force attacks</li> <li>∞ Encryption ...</li> <li>∞ ... Data will be scrambled</li> <li>∞ ... <b>Key</b> is required to decrypt the data</li> <li>∞ ... If data is stolen it will be meaningless</li> <li>∞ Physical security methods ...</li> <li>∞ ... The physical security will need to be overcome</li> <li>∞ ... This can help deter theft of the data</li> <li>∞ Antispyware ...</li> <li>∞ ... will remove any spyware from system</li> <li>∞ ... will prevent data being relayed to a third party</li> </ul> | 6 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

# 0478/13 – NOVEMBER 2019

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 8        | <b>Four from:</b> <ul style="list-style-type: none"> <li>∞ A hacker could have hacked the network ...</li> <li>∞ ... and downloaded the malware onto the network</li> <li>∞ Clicking a link/attachment/downloaded a file from an email/on a webpage ...</li> <li>∞ ... the malware could have been embedded into the link/attachment/file</li> <li>∞ Opening an infected software package ...</li> <li>∞ ... this would trigger the malware to download onto the network</li> <li>∞ Inserting an infected portable storage device ...</li> <li>∞ ... when the drive is accessed the malware is downloaded to the network</li> <li>∞ Firewall has been turned off ...</li> <li>∞ ... so malware would not be detected/checked for when entering network</li> <li>∞ Anti-malware has been turned off ...</li> <li>∞ ... so malware is not detected/checked for when files are downloaded</li> </ul> | 4     |

## 0478/12 – MARCH 2020

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mark |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 8(a)     | Any <b>five</b> from: <ul style="list-style-type: none"><li>• <b>Sends</b> the URL of the website</li><li>• ... to a DNS to find the IP address</li><li>• Connects to the webserver (using the IP address) ...</li><li>• ... using HTTP / HTTPS</li><li>• Renders/Translates the HTML</li><li>• Runs active/client-side scripts built into webpages</li><li>• <b>Manages</b> SSL/TLS certificate process</li><li>• Stores/retrieves cookies</li></ul> | 5    |
| 8(b)     | Any <b>three</b> from: <ul style="list-style-type: none"><li>• <b>Webserver</b> is sent multiple requests // Requests flood the <b>webserver</b> ...</li><li>• ... at the <b>same</b> time</li><li>• <b>Webserver</b> crashes / runs slow</li><li>• Designed to prevent access to e.g. a website // Stops legitimate requests being processed/served</li></ul>                                                                                        | 3    |

## 0478/12 – JUNE 2020

|      |                                                                                                                                                                                                                                                                                       |   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 3(b) | Any <b>three</b> from: <ul style="list-style-type: none"><li>– Firewall</li><li>– Password</li><li>– Proxy server</li><li>– Physical methods (by example e.g. CCTV, Locks)</li><li>– Access rights</li><li>– <u>Asymmetric</u> encryption</li><li>– Disconnect from network</li></ul> | 3 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

| Question  | Answer                                                                                                                                                                                                                                                                                                                                     | Marks |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(a)      | Any <b>three</b> from: <ul style="list-style-type: none"><li>– Convert HTML code</li><li>– Display web pages</li><li>– Check if a website is secure</li><li>– Request web pages from a web server</li><li>– Send URL/domain name</li><li>– Runs active script</li><li>– Store history/favourites/bookmarks</li><li>– Create tabs</li></ul> | 3     |
| 5(b)(i)   | <ul style="list-style-type: none"><li>– Carries out authentication of server and client</li><li>– Handles encryption algorithms / keys</li></ul>                                                                                                                                                                                           | 2     |
| 5(b)(ii)  | <ul style="list-style-type: none"><li>– Record layer</li></ul>                                                                                                                                                                                                                                                                             | 1     |
| 5(b)(iii) | Any <b>one</b> from: <ul style="list-style-type: none"><li>– SSL</li><li>– HTTPS</li></ul>                                                                                                                                                                                                                                                 | 1     |
| 5(c)      | <ul style="list-style-type: none"><li>– Cookies</li></ul>                                                                                                                                                                                                                                                                                  | 1     |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 10(a)     | <b>One</b> mark for similarity, <b>two</b> marks for differences<br>Similarity: <ul style="list-style-type: none"> <li>– Both are designed to steal personal data</li> <li>– They both pose as a real company/person</li> </ul> Differences: <ul style="list-style-type: none"> <li>– Pharming uses malicious code installed on hard drive</li> <li>– Phishing is in form of an email</li> <li>– Phishing requires use to follow a link / open an attachment</li> </ul> | 3     |
| 10(b)     | <ul style="list-style-type: none"> <li>– Virus</li> <li>– Malware</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| 10(c)(i)  | <ul style="list-style-type: none"> <li>– Incorrect</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                           | 1     |
| 10(c)(ii) | Any <b>four</b> from: <ul style="list-style-type: none"> <li>– Can help prevent hacking</li> <li>– Can monitor incoming and outgoing traffic</li> <li>– Can set criteria / rules are set for traffic</li> <li>– Can check whether traffic meets / defies criteria rules</li> <li>– Can rejects any traffic that does not meet / defies criteria</li> </ul>                                                                                                              | 4     |

## 0478/13 – JUNE 2020

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3(a)     | Any <b>three</b> from: <ul style="list-style-type: none"> <li>– Sends request to webserver</li> <li>– Receives web pages back from webserver</li> <li>– Converts HTML to display web page</li> <li>– Manages protocols</li> </ul>                                                                                                                                                    | 3     |
| 3(b)     | Any <b>three</b> from: <ul style="list-style-type: none"> <li>– <b>Many</b> requests are sent from a computer</li> <li>– Requests are sent to the <u>webserver</u></li> <li>– The webserver becomes flooded with traffic</li> <li>– The webserver cannot handle the requests / fails</li> <li>– The website can no longer be accessed</li> <li>– Attack maybe distributed</li> </ul> | 3     |

| Question | Answer                                                                                                                                                          | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4        | <ul style="list-style-type: none"> <li>– Serial (ignore any ref to simplex etc.)</li> <li>– IP (address)</li> <li>– Browser</li> <li>– MAC (address)</li> </ul> | 4     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(a)     | Any <b>four</b> from: <ul style="list-style-type: none"> <li>– Examines outgoing traffic to check what is being requested</li> <li>– Examines incoming traffic to check the content of what is being received</li> <li>– Sets rules/criteria for websites that can/cannot be accessed // creates a blacklist</li> <li>– Check if traffic meets/does not meet rules/criteria</li> <li>– If it does/does not, access to website granted/denied</li> </ul> | 4     |
| 7(b)     | Any <b>three</b> from: <ul style="list-style-type: none"> <li>– Software that can replicate itself</li> <li>– It could cause the computer to crash / run slow / generate errors</li> <li>– It could delete/damage files</li> <li>– It could fill up the storage space</li> <li>– It could stop the hardware being able to communicate</li> <li>– It could spread to other devices on the network</li> </ul>                                             | 3     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(c)(i)  | Any <b>two</b> from: <ul style="list-style-type: none"> <li>– Locked padlock</li> <li>– HTTPS</li> <li>– View the certificate</li> </ul>                                                                                                                                                                                                                                                                                                                              | 2     |
| 7(c)(ii) | Any <b>four</b> from: <ul style="list-style-type: none"> <li>– requests web server to identify itself/view the (SSL) certificate</li> <li>– receives a copy of the (SSL) certificate, sent from the webserver</li> <li>– checks if (SSL) certificate is authentic/trustworthy</li> <li>– sends signal back to webserver that the certificate is authentic/trustworthy</li> <li>– starts to transmit data once connection is established as secure</li> </ul>          | 4     |
| 10(a)    | Any <b>four</b> from: <ul style="list-style-type: none"> <li>– Example of spyware e.g. Keylogger is used</li> <li>– Spyware is downloaded without knowledge (by example)</li> <li>– Spyware records key presses / screen clicks / screen activity</li> <li>– Data is relayed back to third party</li> <li>– Data is analysed // Patterns in data could reveal log-in details ...</li> <li>– ... details can then be used to log into the laptop (remotely)</li> </ul> | 4     |
| 10(b)    | <ul style="list-style-type: none"> <li>– Biometric device</li> <li>– Two step authentication</li> </ul>                                                                                                                                                                                                                                                                                                                                                               | 1     |

## 0478/11 – NOVEMBER 2020

| Question | Answer                                                                                                                                          | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3(a)     | <ul style="list-style-type: none"> <li>– Handshake</li> <li>– Record</li> </ul>                                                                 | 2     |
| 3(b)     | <ul style="list-style-type: none"> <li>– Web server</li> <li>– Certificate</li> <li>– Authentic</li> <li>– Browser</li> <li>– Signal</li> </ul> | 5     |
| 3(c)     | Any <b>one</b> from: <ul style="list-style-type: none"> <li>– SSL</li> <li>– HTTPS</li> </ul>                                                   | 1     |
| 5(b)     | <ul style="list-style-type: none"> <li>– Firewall</li> <li>– Proxy server</li> </ul>                                                            | 2     |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(d)(i)  | <ul style="list-style-type: none"> <li>– Handshake (layer)</li> <li>– Record (layer)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2 |
| 1(d)(ii) | Any <b>six</b> from: <ul style="list-style-type: none"> <li>– <b>Client/browser</b> requests secure <b>connection</b> to server</li> <li>– <b>Client/browser</b> requests the <b>server</b> to identify itself</li> <li>– <b>Server</b> provides a digital certificate</li> <li>– <b>Client/browser</b> validates the certificate</li> <li>– <b>Client/browser</b> send signal <b>back to server</b> (to begin transmission)</li> <li>– Session caching can be used</li> <li>– A session key is generated</li> <li>– Encryption method is agreed // data is encrypted</li> </ul> | 6 |
| 1(e)(i)  | Any <b>three</b> from: <ul style="list-style-type: none"> <li>– Hacking</li> <li>– Denial of service (DoS) attack</li> <li>– Virus</li> <li>– Malware</li> </ul> <p>NOTE: Three different type of malware can be awarded</p>                                                                                                                                                                                                                                                                                                                                                     | 3 |
| 1(e)(ii) | Any <b>four</b> from: <ul style="list-style-type: none"> <li>– Acts as a firewall</li> <li>– Monitor/filters/examines incoming <b>and</b> outgoing traffic</li> <li>– Rules/criteria for traffic can be <b>set</b> // blacklist/whitelist <b>set</b></li> <li>– Blocks any traffic that does not meet criteria ...</li> <li>– ... and can send a warning message <b>to the user</b></li> <li>– Stop the website failing in a DoS attack // DoS attack hits the proxy server and not the webserver</li> </ul>                                                                     | 4 |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(c)     | Any <b>four</b> from: <ul style="list-style-type: none"> <li>– Allows user to view web pages</li> <li>– Renders HTML</li> <li>– Allows user to bookmark/favourite web pages</li> <li>– Provides navigation features</li> <li>– Allows (multiple) tabs</li> <li>– <b>Stores</b> cookies</li> <li>– Records history of pages visited</li> <li>– Has a homepage</li> <li>– Runs active script</li> <li>– Allows <b>files</b> to be downloaded from <b>website/internet</b></li> <li>– Sends a request to the <b>IP address/web server</b> (to obtain the contents of a web page)</li> <li>– Sends URL to DNS</li> <li>– <b>Manages</b> HTTP/HTTPS protocol</li> </ul> | 4     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4(a)     | Any <b>four</b> from: <ul style="list-style-type: none"><li>– Browsers sends URL to DNS</li><li>– ... using HTTP</li><li>– DNS finds matching IP addresses for URL</li><li>– ... and sends IP address to web browser</li><li>– Web browser sends request to IP address/web server for web pages</li><li>– Web pages are sent from web server to browser</li><li>– Browser renders HTML to display web pages</li><li>– Any security certificates are exchanged/authenticated // SSL/HTTPS is used to secure the data</li><li>– ... encrypting any data sent</li></ul> | 4     |

| Question | Answer                                                                                                                                                                                                           | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4(b)     | Any <b>three</b> from: <ul style="list-style-type: none"><li>– Hacking</li><li>– Denial of service (DoS)</li><li>– Malware</li><li>– Virus</li></ul> <p>NOTE: three suitable types of malware can be awarded</p> | 3     |

| Question | Answer                                                                                                   | Marks |
|----------|----------------------------------------------------------------------------------------------------------|-------|
| 5(c)     | Any <b>two</b> from: <ul style="list-style-type: none"><li>– TLS</li><li>– SSL</li><li>– HTTPS</li></ul> | 2     |

**PUBLISHED**

| Question | Answer                                                                                                                                                                                                                                      | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 13(a)    | Any <b>one</b> from: <ul style="list-style-type: none"><li>– Both are designed to steal/collect personal data</li><li>– Both pretend to be a real company</li><li>– Both use fake websites</li></ul>                                        | 1     |
| 13(b)    | <ul style="list-style-type: none"><li>– Phishing involves use of an email whereas pharming involves installing malicious code</li><li>– Phishing involves clicking a link or an attachment whereas pharming creates a redirection</li></ul> | 2     |



## 0478/11 – MARCH 2021

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(c)(i)  | – Data if intercepted cannot be understood // Data is <b>encrypted</b> // Data is <b>scrambled</b> // uses keys to <b>encode/decode</b> data                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>1</b> |
| 2(c)(ii) | <b>Four from:</b> <ul style="list-style-type: none"><li>– Uses (digital) <b>certificates</b></li><li>– ....requested from web server by browser/client // browser/client asks web server to identify itself</li><li>– Server send SSL/digital signature to browser/client</li><li>– Client and server agree on encryption method to use</li><li>– ... that contains the server's public key</li><li>– Browser checks <b>authenticity of certificate</b>...</li><li>– ... then session key is generated</li><li>– ... the transaction will begin // sends signal to server to start transmission</li></ul> | <b>4</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6(a)     | Any <b>four</b> from: <ul style="list-style-type: none"><li>– Monitors incoming and outgoing traffic</li><li>– Allows the <b>setting</b> of criteria/blacklist/whitelist/by example</li><li>– Blocks access to signals that do not meet requirements/criteria/blacklist/whitelist ...</li><li>– ... sends signal to warn the user</li><li>– Restrict access to specific applications</li><li>– Blocks entry/exit by specific ports</li></ul>                                                                                                                                                                 | <b>4</b> |
| 6(b)     | <b>One</b> mark for risk, <b>two</b> marks for description <ul style="list-style-type: none"><li>– Phishing</li><li>– <b>Legitimate looking email</b> sent to user</li><li>– Clicking on <b>link/attachment</b> takes user to fake website</li><li>– Pharming</li><li>– Software is installed on user's computer</li><li>– Redirects (correct URL) to different/fraudulent website</li><li>– Spyware (accept keylogger but do not award for MP3)</li><li>– Software is installed on user's computer</li><li>– Records key strokes // keylogger</li><li>– Transmits data to third part for analysis</li></ul> | <b>6</b> |

## 0478/11 – JUNE 2021

| Question | Answer                                                                                                                                                                                                                       | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(a)     | One mark per each correct term in the correct order. <ul style="list-style-type: none"><li>– Software</li><li>– Network</li><li>– Criteria</li><li>– Accept // reject</li><li>– Reject // accept</li><li>– Hacking</li></ul> | <b>6</b> |



| Question | Answer                                                                                                                                                                                                                                                                  | Marks    |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(b)     | Any <b>three</b> from: <ul style="list-style-type: none"> <li>– Password</li> <li>– Biometrics (device)</li> <li>– Encryption</li> <li>– Physical methods (e.g. locks)</li> <li>– Two-factor authentication // Two-step verification</li> <li>– Anti-viruses</li> </ul> | <b>3</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4        | Any <b>six</b> from: <p>Phishing</p> <ul style="list-style-type: none"> <li>– Legitimate looking email sent to user</li> <li>– encourages user to <b>click a link</b> that directs user to a fake website</li> <li>– User encouraged to enter personal details into a fake website // designed to obtain personal details from a user</li> </ul> <p>Pharming</p> <ul style="list-style-type: none"> <li>– Malicious code/malware is downloaded/installed // software downloaded without users' knowledge</li> <li>– ... that <b>re-directs</b> user to fake website (when legitimate URL entered)</li> <li>– User encouraged to enter personal details into a fake website // designed to obtain personal details from a user</li> </ul> | <b>6</b> |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(b) | Any <b>four</b> from: <ul style="list-style-type: none"> <li>– Browser sends URL to DNS</li> <li>– ... using HTTP/HTTPS</li> <li>– IP address is found on DNS</li> <li>– DNS returns IP address to the browser</li> <li>– Browser sends request to web server/IP address</li> <li>– Web server sends web pages back to browser</li> <li>– Browser <b>interprets/renders</b> the HTML (to display web pages)</li> <li>– Security certificates exchanged</li> </ul> | <b>4</b> |
| 7(c) | <b>Two</b> from: <ul style="list-style-type: none"> <li>– <b>Web server</b> has been flooded with traffic // <b>web server</b> has been sent many requests at once</li> <li>– ... so, server is brought to a halt / crashes</li> </ul>                                                                                                                                                                                                                            | <b>2</b> |

| Question                                 | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |              |                |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------|----------------|-----------------------------|-----------------------------|--------------------------------------------|--|---|--|------------------------------------|---|---|--|------------------------------|--|--|---|---------------------------------|---|---|--|---------------------|---|--|--|------------------------------------------|---|--|--|
| 4(a)                                     | One mark per each correct row.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6     |              |                |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
|                                          | <table><tr><th>Statement</th><th>Virus<br/>(✓)</th><th>Spyware<br/>(✓)</th><th>Denial of<br/>service<br/>(✓)</th></tr><tr><td>captures all data entered using a keyboard</td><td></td><td>✓</td><td></td></tr><tr><td>can be installed onto a web server</td><td>✓</td><td>✓</td><td></td></tr><tr><td>prevents access to a website</td><td></td><td></td><td>✓</td></tr><tr><td>is malicious code on a computer</td><td>✓</td><td>✓</td><td></td></tr><tr><td>is self-replicating</td><td>✓</td><td></td><td></td></tr><tr><td>damages the files on a user's hard drive</td><td>✓</td><td></td><td></td></tr></table> |       | Statement    | Virus<br>(✓)   | Spyware<br>(✓)              | Denial of<br>service<br>(✓) | captures all data entered using a keyboard |  | ✓ |  | can be installed onto a web server | ✓ | ✓ |  | prevents access to a website |  |  | ✓ | is malicious code on a computer | ✓ | ✓ |  | is self-replicating | ✓ |  |  | damages the files on a user's hard drive | ✓ |  |  |
|                                          | Statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       | Virus<br>(✓) | Spyware<br>(✓) | Denial of<br>service<br>(✓) |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
|                                          | captures all data entered using a keyboard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |              | ✓              |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
|                                          | can be installed onto a web server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       | ✓            | ✓              |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
|                                          | prevents access to a website                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |              |                | ✓                           |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
|                                          | is malicious code on a computer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       | ✓            | ✓              |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
|                                          | is self-replicating                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       | ✓            |                |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
| damages the files on a user's hard drive | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |              |                |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
| 4(b)                                     | Any <b>three</b> from:<br>– Phishing<br>– Pharming<br>– Hacking // cracking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3     |              |                |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |
| 4(c)                                     | Any <b>three</b> from:<br>– Human error<br>– Power failure/surge<br>– Hardware failure<br>– Software failure<br>– Fire<br>– Flood                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3     |              |                |                             |                             |                                            |  |   |  |                                    |   |   |  |                              |  |  |   |                                 |   |   |  |                     |   |  |  |                                          |   |  |  |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6(a) | <p>Any <b>three</b> from:</p> <ul style="list-style-type: none"> <li>– Webserver sends (cookie) file to user's <b>browser</b></li> <li>– User's payment details stored in <b>encrypted</b> text file // data is encrypted to be stored</li> <li>– Cookie file is stored by <b>browser/on user's HDD/SSD</b></li> <li>– When user revisits website, <b>webserver</b> requests cookie file // <b>webserver</b> can access the data stored in the cookie file (to automatically enter details)</li> <li>– ... and <b>browser</b> sends cookie file back to <b>webserver</b> (to automatically enter the details)</li> </ul>                                                                                                          | <b>3</b> |
| 6(b) | <p><b>Four</b> from:</p> <ul style="list-style-type: none"> <li>– User does not see what information is stored // might collect data that user does not know about ...</li> <li>– ... so, user may feel their privacy is affected</li> <li>– A profile could be built about the user ...</li> <li>– ... that could expose a user's identity // lead to identity theft</li> <li>– Sensitive information stored in cookies could be intercepted in transmission ...</li> <li>– Other websites could gain access to the cookies stored on a user's computer ...</li> <li>– Computer could be hacked to obtain data stored in cookies ...</li> <li>– ... so, payment information could be stolen and used by a third party</li> </ul> | <b>4</b> |

**0478/13 – JUNE 2021**

| Question | Answer                                                                                                                                                                                                                                  | Marks    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a)     | <ul style="list-style-type: none"> <li>– Legitimate looking/fake email sent to user</li> <li>– ... that contains a link to a fake website</li> <li>– User <b>clicks link</b> and enters personal details (into fake website)</li> </ul> | <b>3</b> |
| 4(b)     | <p>Any <b>two</b> from:</p> <ul style="list-style-type: none"> <li>– Pharming</li> <li>– Spyware</li> <li>– Hacking/cracking</li> </ul>                                                                                                 | <b>2</b> |

## NOVEMBER 2021 (1)

| Question                         | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |             |      |                                        |         |                                                             |            |                                                                                               |        |                                                                                 |                                  |                                                        |   |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|------|----------------------------------------|---------|-------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------|---|
| 11                               | <p>One mark per each correct term.</p> <table><tr><th>Terms</th><th>Description</th></tr><tr><td>HTML</td><td>the language used to create a web page</td></tr><tr><td>Browser</td><td>the type of software application used to display a web page</td></tr><tr><td>IP address</td><td>an address given to a computer, by a network, to allow the computer to be uniquely identified</td></tr><tr><td>Cookie</td><td>a text file sent by a web server to collect data about a user's browsing habits</td></tr><tr><td>Internet Service Provider // ISP</td><td>the company that provides a connection to the Internet</td></tr></table> | Terms | Description | HTML | the language used to create a web page | Browser | the type of software application used to display a web page | IP address | an address given to a computer, by a network, to allow the computer to be uniquely identified | Cookie | a text file sent by a web server to collect data about a user's browsing habits | Internet Service Provider // ISP | the company that provides a connection to the Internet | 5 |
| Terms                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |      |                                        |         |                                                             |            |                                                                                               |        |                                                                                 |                                  |                                                        |   |
| HTML                             | the language used to create a web page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |             |      |                                        |         |                                                             |            |                                                                                               |        |                                                                                 |                                  |                                                        |   |
| Browser                          | the type of software application used to display a web page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |      |                                        |         |                                                             |            |                                                                                               |        |                                                                                 |                                  |                                                        |   |
| IP address                       | an address given to a computer, by a network, to allow the computer to be uniquely identified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |             |      |                                        |         |                                                             |            |                                                                                               |        |                                                                                 |                                  |                                                        |   |
| Cookie                           | a text file sent by a web server to collect data about a user's browsing habits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |             |      |                                        |         |                                                             |            |                                                                                               |        |                                                                                 |                                  |                                                        |   |
| Internet Service Provider // ISP | the company that provides a connection to the Internet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |             |      |                                        |         |                                                             |            |                                                                                               |        |                                                                                 |                                  |                                                        |   |

## NOVEMBER 2021 (2)

| Question | Answer                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(a)     | <p>Any <b>three</b> from:</p> <ul style="list-style-type: none"> <li>- Password</li> <li>- Add a biometric device to the laptop // set biometric password</li> <li>- Use two-step verification // Use two factor authentication</li> <li>- Physically lock the laptop away in a secure cupboard // Taking laptop with him at all times</li> </ul> | 3     |

| Question | Answer                                                                                                                                                                                                                                                                     | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 10(a)    | <ul style="list-style-type: none"> <li>- Enables an encrypted link (between the browser and the web server) // It encrypts the data</li> <li>- ... based on the authentication of an (SSL) certificate // and will only send it if the certificate is authentic</li> </ul> | 2     |
| 10(b)    | <ul style="list-style-type: none"> <li>- Transport Layer Security // TLS</li> </ul>                                                                                                                                                                                        | 1     |
| 10(c)    | <p>Any <b>two</b> from:</p> <ul style="list-style-type: none"> <li>- URL begins with HTTPS</li> <li>- Padlock symbol is locked</li> <li>- Check the certificate is valid</li> </ul>                                                                                        | 2     |

|      |                                                                                                                                 |          |
|------|---------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(d) | Any <b>two</b> from: <ul style="list-style-type: none"><li>– URL begins with HTTPS</li><li>– Padlock symbol is locked</li></ul> | <b>2</b> |
|------|---------------------------------------------------------------------------------------------------------------------------------|----------|

| Question | Answer                                                                                                                                           | Marks    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10(a)    | – To disrupt the operation of a web server/network                                                                                               | <b>1</b> |
| 10(b)    | – To obtain a user's <b>personal</b> data                                                                                                        | <b>1</b> |
| 10(c)    | Any <b>three</b> from: <ul style="list-style-type: none"><li>– Hacking // Cracking</li><li>– Virus</li><li>– Spyware</li><li>– Malware</li></ul> | <b>3</b> |

| Question        | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks           |          |     |                                                                                                                                                                                                                 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 5(b)            | <div>1 mark for each bullet to max 2 for each</div> <table><tr><th>Type of address</th><th>Features</th></tr><tr><td>URL</td><td><ul style="list-style-type: none"><li>• protocol</li><li>• domain name</li><li>• top level domain</li><li>• file/folder name/address/directory/path</li><li>• made up of letters, numbers and symbols</li></ul></td></tr><tr><td>IP</td><td><p>IPv4:</p><ul style="list-style-type: none"><li>• 4 groups of numbers // 12 numbers</li><li>• separated by full stops (.)</li><li>• 0-255 in each group</li><li>• 4 bytes long // 32 bits</li><li>• Represented in denary</li></ul><p>IPv6:</p><ul style="list-style-type: none"><li>• 8 groups of numbers // 32 numbers</li><li>• separated by colons (:)</li><li>• 0-FFFF in each group</li><li>• 16 bytes long // 128 bits</li><li>• Consecutive groups of 0000 can be replaced with ::</li><li>• Represented in hexadecimal</li></ul></td></tr><tr><td>MAC</td><td><ul style="list-style-type: none"><li>• First part is manufacturer ID</li><li>• Second part is serial number</li><li>• Each part has 3 pairs of numbers // Each part has 6 numbers // 6 pairs of 2-digits // consists of 12 numbers</li><li>• ...between 00 and FF</li><li>• ..separated by :</li><li>• Each part is 3 bytes // Is 6 bytes long // Is 48 bits long</li><li>• Represented using hexadecimal</li></ul></td></tr></table> | Type of address | Features | URL | <ul style="list-style-type: none"><li>• protocol</li><li>• domain name</li><li>• top level domain</li><li>• file/folder name/address/directory/path</li><li>• made up of letters, numbers and symbols</li></ul> | IP | <p>IPv4:</p> <ul style="list-style-type: none"><li>• 4 groups of numbers // 12 numbers</li><li>• separated by full stops (.)</li><li>• 0-255 in each group</li><li>• 4 bytes long // 32 bits</li><li>• Represented in denary</li></ul> <p>IPv6:</p> <ul style="list-style-type: none"><li>• 8 groups of numbers // 32 numbers</li><li>• separated by colons (:)</li><li>• 0-FFFF in each group</li><li>• 16 bytes long // 128 bits</li><li>• Consecutive groups of 0000 can be replaced with ::</li><li>• Represented in hexadecimal</li></ul> | MAC | <ul style="list-style-type: none"><li>• First part is manufacturer ID</li><li>• Second part is serial number</li><li>• Each part has 3 pairs of numbers // Each part has 6 numbers // 6 pairs of 2-digits // consists of 12 numbers</li><li>• ...between 00 and FF</li><li>• ..separated by :</li><li>• Each part is 3 bytes // Is 6 bytes long // Is 48 bits long</li><li>• Represented using hexadecimal</li></ul> | 6 |
| Type of address | Features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |          |     |                                                                                                                                                                                                                 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| URL             | <ul style="list-style-type: none"><li>• protocol</li><li>• domain name</li><li>• top level domain</li><li>• file/folder name/address/directory/path</li><li>• made up of letters, numbers and symbols</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |          |     |                                                                                                                                                                                                                 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| IP              | <p>IPv4:</p> <ul style="list-style-type: none"><li>• 4 groups of numbers // 12 numbers</li><li>• separated by full stops (.)</li><li>• 0-255 in each group</li><li>• 4 bytes long // 32 bits</li><li>• Represented in denary</li></ul> <p>IPv6:</p> <ul style="list-style-type: none"><li>• 8 groups of numbers // 32 numbers</li><li>• separated by colons (:)</li><li>• 0-FFFF in each group</li><li>• 16 bytes long // 128 bits</li><li>• Consecutive groups of 0000 can be replaced with ::</li><li>• Represented in hexadecimal</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |          |     |                                                                                                                                                                                                                 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| MAC             | <ul style="list-style-type: none"><li>• First part is manufacturer ID</li><li>• Second part is serial number</li><li>• Each part has 3 pairs of numbers // Each part has 6 numbers // 6 pairs of 2-digits // consists of 12 numbers</li><li>• ...between 00 and FF</li><li>• ..separated by :</li><li>• Each part is 3 bytes // Is 6 bytes long // Is 48 bits long</li><li>• Represented using hexadecimal</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |          |     |                                                                                                                                                                                                                 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                                                                                                                                                                                                                                                                                                                                                                      |   |

| Question                      | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks          |                   |          |                   |                                                                                                                                                    |                |               |                                                                                                                                                                                       |            |                               |                                                                                        |                |   |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|----------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------|----------------------------------------------------------------------------------------|----------------|---|
| 5(c)                          | <p>1 mark for threat. 1 for impact. 1 for software.</p> <p>Do not award identical impacts twice but read whole answer and award if additional impact given. Allow the same software twice. e.g.</p> <table border="1"> <thead> <tr> <th>Threat</th><th>Impact on company</th><th>Software</th></tr> </thead> <tbody> <tr> <td>Denial of service</td><td> <ul style="list-style-type: none"> <li>Users cannot access the website</li> <li>Loss of sales (of holidays)</li> <li>Loss of reputation</li> </ul> </td><td>Proxy/firewall</td></tr> <tr> <td>Virus/malware</td><td> <ul style="list-style-type: none"> <li>Data on the server may be deleted/changed</li> <li>Website may be deleted/changed</li> <li>Server may be filled with data and crash</li> </ul> </td><td>Anti-virus</td></tr> <tr> <td>Unauthorised access // hacker</td><td> <ul style="list-style-type: none"> <li>Data could be deleted/stolen/changed</li> </ul> </td><td>Proxy/Firewall</td></tr> </tbody> </table> | Threat         | Impact on company | Software | Denial of service | <ul style="list-style-type: none"> <li>Users cannot access the website</li> <li>Loss of sales (of holidays)</li> <li>Loss of reputation</li> </ul> | Proxy/firewall | Virus/malware | <ul style="list-style-type: none"> <li>Data on the server may be deleted/changed</li> <li>Website may be deleted/changed</li> <li>Server may be filled with data and crash</li> </ul> | Anti-virus | Unauthorised access // hacker | <ul style="list-style-type: none"> <li>Data could be deleted/stolen/changed</li> </ul> | Proxy/Firewall | 6 |
| Threat                        | Impact on company                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Software       |                   |          |                   |                                                                                                                                                    |                |               |                                                                                                                                                                                       |            |                               |                                                                                        |                |   |
| Denial of service             | <ul style="list-style-type: none"> <li>Users cannot access the website</li> <li>Loss of sales (of holidays)</li> <li>Loss of reputation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Proxy/firewall |                   |          |                   |                                                                                                                                                    |                |               |                                                                                                                                                                                       |            |                               |                                                                                        |                |   |
| Virus/malware                 | <ul style="list-style-type: none"> <li>Data on the server may be deleted/changed</li> <li>Website may be deleted/changed</li> <li>Server may be filled with data and crash</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Anti-virus     |                   |          |                   |                                                                                                                                                    |                |               |                                                                                                                                                                                       |            |                               |                                                                                        |                |   |
| Unauthorised access // hacker | <ul style="list-style-type: none"> <li>Data could be deleted/stolen/changed</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Proxy/Firewall |                   |          |                   |                                                                                                                                                    |                |               |                                                                                                                                                                                       |            |                               |                                                                                        |                |   |

| Question | Answer                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(a)     | Any <b>one</b> from: <ul style="list-style-type: none"> <li>• Uses biological data</li> <li>• It uses <b>characteristics/features</b> that belong to a <b>human</b></li> </ul>                                                                                                                                                   | 1     |
| 7(b)     | Any <b>two</b> from: <ul style="list-style-type: none"> <li>• A biometric password cannot be guessed</li> <li>• It is very difficult to fake a biometric password</li> <li>• A biometric password cannot be recorded by a keylogger/spyware</li> <li>• A perpetrator cannot shoulder surf to see a biometric password</li> </ul> | 2     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 7(c)     | Any <b>three</b> from: <ul style="list-style-type: none"> <li>• The traffic/data is <b>compared to</b> set criteria/rules/whitelist/blacklist</li> <li>• If the traffic/data does/does not meet the criteria/rules/whitelist/blacklist it will be rejected/blocked ...</li> <li>• ... and an alert can be sent to warn Adele</li> <li>• ... this can help prevent hackers/malicious software (gaining access to the data)</li> </ul> | 3     |
| 7(d)     | <ul style="list-style-type: none"> <li>• It makes the data meaningless</li> </ul>                                                                                                                                                                                                                                                                                                                                                    | 1     |

| Question                                                          | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks             |                |                   |            |                             |  |  |   |                          |   |  |  |                                                                   |   |  |  |                         |  |   |   |                   |   |  |  |   |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------|-------------------|------------|-----------------------------|--|--|---|--------------------------|---|--|--|-------------------------------------------------------------------|---|--|--|-------------------------|--|---|---|-------------------|---|--|--|---|
| 9(a)                                                              | <p>One mark for each correct row:</p> <table><tr><th>Statement</th><th>Browser<br/>(✓)</th><th>IP address<br/>(✓)</th><th>URL<br/>(✓)</th></tr><tr><td>it contains the domain name</td><td></td><td></td><td>✓</td></tr><tr><td>it is a type of software</td><td>✓</td><td></td><td></td></tr><tr><td>it converts Hypertext Markup Language (HTML) to display web pages</td><td>✓</td><td></td><td></td></tr><tr><td>it is a type of address</td><td></td><td>✓</td><td>✓</td></tr><tr><td>it stores cookies</td><td>✓</td><td></td><td></td></tr></table> | Statement         | Browser<br>(✓) | IP address<br>(✓) | URL<br>(✓) | it contains the domain name |  |  | ✓ | it is a type of software | ✓ |  |  | it converts Hypertext Markup Language (HTML) to display web pages | ✓ |  |  | it is a type of address |  | ✓ | ✓ | it stores cookies | ✓ |  |  | 5 |
| Statement                                                         | Browser<br>(✓)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | IP address<br>(✓) | URL<br>(✓)     |                   |            |                             |  |  |   |                          |   |  |  |                                                                   |   |  |  |                         |  |   |   |                   |   |  |  |   |
| it contains the domain name                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   | ✓              |                   |            |                             |  |  |   |                          |   |  |  |                                                                   |   |  |  |                         |  |   |   |                   |   |  |  |   |
| it is a type of software                                          | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                |                   |            |                             |  |  |   |                          |   |  |  |                                                                   |   |  |  |                         |  |   |   |                   |   |  |  |   |
| it converts Hypertext Markup Language (HTML) to display web pages | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                |                   |            |                             |  |  |   |                          |   |  |  |                                                                   |   |  |  |                         |  |   |   |                   |   |  |  |   |
| it is a type of address                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ✓                 | ✓              |                   |            |                             |  |  |   |                          |   |  |  |                                                                   |   |  |  |                         |  |   |   |                   |   |  |  |   |
| it stores cookies                                                 | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                |                   |            |                             |  |  |   |                          |   |  |  |                                                                   |   |  |  |                         |  |   |   |                   |   |  |  |   |

| Question                         | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|---------|-------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------|--------------------------------|----------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------|---|
| 3(a)                             | <p><b>One mark for the correct term or definition</b></p> <table><tr><th>Term</th><th>Definition</th></tr><tr><td>browser</td><td><b>Software/application</b> that allows users to view web pages / render HTML</td></tr><tr><td>Internet Service provider // ISP</td><td>this is the company that provides a user with a connection to the Internet</td></tr><tr><td>HTTP // HTTPS</td><td>this is a protocol that is used to send data for web pages across the Internet</td></tr><tr><td>Uniform Resource Locator (URL)</td><td>a <b>text-based</b> version of a web address</td></tr><tr><td>cookie</td><td>a text <b>file</b> (stored by web browser) that contains data about a <b>user's browsing habits/details/preferences</b></td></tr></table> | Term  | Definition | browser | <b>Software/application</b> that allows users to view web pages / render HTML | Internet Service provider // ISP | this is the company that provides a user with a connection to the Internet | HTTP // HTTPS | this is a protocol that is used to send data for web pages across the Internet | Uniform Resource Locator (URL) | a <b>text-based</b> version of a web address | cookie | a text <b>file</b> (stored by web browser) that contains data about a <b>user's browsing habits/details/preferences</b> | 5 |
| Term                             | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |
| browser                          | <b>Software/application</b> that allows users to view web pages / render HTML                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |
| Internet Service provider // ISP | this is the company that provides a user with a connection to the Internet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |
| HTTP // HTTPS                    | this is a protocol that is used to send data for web pages across the Internet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |
| Uniform Resource Locator (URL)   | a <b>text-based</b> version of a web address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |
| cookie                           | a text <b>file</b> (stored by web browser) that contains data about a <b>user's browsing habits/details/preferences</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |
| 3(b)                             | <ul style="list-style-type: none"><li>• Hardware or software based</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1     |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |
| 3(c)                             | <p><b>Four from:</b></p> <ul style="list-style-type: none"><li>• (The parent can) set criteria for the websites she is allowed to visit</li><li>• ... such as a whitelist/blacklist of websites</li><li>• The firewall will examine the data/traffic <b>incoming</b> and <b>outgoing</b> from her computer</li><li>• If data is sent from a website that is not allowed, it will be blocked</li></ul>                                                                                                                                                                                                                                                                                                                                                     | 4     |            |         |                                                                               |                                  |                                                                            |               |                                                                                |                                |                                              |        |                                                                                                                         |   |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 6(a)     | <p><b>One mark for identifying the attack, two marks for the description</b></p> <ul style="list-style-type: none"> <li>Phishing</li> <li>Email is sent to user to encourage them to click link</li> <li>... that takes user to fake website</li> </ul> <p>Pharming</p> <ul style="list-style-type: none"> <li>Email is sent to user to encourage them to click link/download attachment</li> <li>... that triggers <b>download</b> of <b>malicious code</b> that will redirect user to fake website</li> </ul> <p>Virus/malware</p> <ul style="list-style-type: none"> <li>Email is sent to user to encourage them to click link/download attachment</li> <li>... that triggers <b>download</b> of virus/malware</li> </ul> <p>Denial of service // DoS</p> <ul style="list-style-type: none"> <li>A very large number of emails are sent to a <b>server/network at the same time</b></li> <li>... crashing the <b>server/network</b></li> </ul> | 6     |
| 6(b)     | <p>Any <b>two</b> from:</p> <ul style="list-style-type: none"> <li>Encryption</li> <li>Password</li> <li>Two-step / Two-factor authentication/verification</li> <li>Biometric device</li> <li>Anti-malware // Anti-virus</li> <li>Proxy-server</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |



# JUNE 2022(3)

| Question | Answer                                                                                                                                                                 | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a)     | <b>Three</b> from: <ul style="list-style-type: none"><li>• Protocol</li><li>• Domain name / Web server name</li><li>• Filename / web page name / folder name</li></ul> | <b>3</b> |

| Question | Answer                                                                                                                                                                 | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(b)     | <ul style="list-style-type: none"><li>• It is sent to a DNS ...</li><li>• which looks up the <b>corresponding/matching IP</b> address</li></ul>                        | <b>2</b> |
| 4(c)(i)  | <ul style="list-style-type: none"><li>• HTTPS</li></ul>                                                                                                                | <b>1</b> |
| 4(c)(ii) | Any <b>one</b> from: <ul style="list-style-type: none"><li>• It encrypts it</li><li>• It applies encryption algorithm</li><li>• It applies an encryption key</li></ul> | <b>1</b> |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(a) | D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1 |
| 1(b) | <p><b>One</b> mark for identification. E.g. <b>One</b> mark per bullet for description to <b>max two</b> each.</p> <p>Virus</p> <ul style="list-style-type: none"> <li>• Software/code that replicates</li> <li>• ...when the user runs it // with an active host</li> <li>• Deletes/damages/corrupts data/files // takes up storage/memory space</li> </ul> <p>Worm</p> <ul style="list-style-type: none"> <li>• Software/code that replicates itself on a <b>network</b></li> <li>• ...without user input // without active host</li> <li>• Takes-up bandwidth</li> <li>• Deletes/damages/corrupts data/files // takes up storage/memory space</li> <li>• Opens back doors to computers over the network</li> <li>• Used to deposit other malware on networked computers</li> </ul> <p>Trojan horse</p> <ul style="list-style-type: none"> <li>• Software/code that is hidden within other software // Software that is disguised as authentic software</li> <li>• ...when <b>downloaded/installed</b> the other <b>malware/by example</b> it contains is <b>installed</b></li> </ul> <p>Adware</p> <ul style="list-style-type: none"> <li>• Software/code that generates/displays (unwanted) adverts on a user's computer</li> <li>• Some may contain spyware/other malware</li> <li>• Some when clicked may link to viruses</li> <li>• Reduces device performance // reduces internet speed</li> <li>• Redirects internet searches/user to fake websites</li> </ul> <p>Ransomware</p> <ul style="list-style-type: none"> <li>• Software/code that stops a user accessing/using their computer/data</li> <li>• ...by encrypting the data/files/computer</li> <li>• A <b>fee</b> has to be paid to decrypt the data // A <b>fee</b> has to be paid to 'release' the computer/device/data</li> </ul> | 6 |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(c)     | <p><b>One</b> mark for each similarity to <b>max two</b>. <b>One</b> mark for difference (both sides needed unless clearly and accurately implied).</p> <p>Similarities e.g.</p> <ul style="list-style-type: none"> <li>• Check incoming and outgoing signals // filter traffic</li> <li>• Store whitelist/blacklist</li> <li>• Block incoming/outgoing signals</li> <li>• Both block unauthorised access</li> <li>• Keep a log of traffic</li> <li>• Both can be hardware or software (or both)</li> </ul> <p>Differences e.g.</p> <ul style="list-style-type: none"> <li>• Proxy can hide user's IP address, firewall does not hide the user's IP address</li> <li>• Proxy intention is to divert attack from server, firewall is to stop unauthorised access</li> <li>• Proxy protects a server, firewall protects individual computer</li> <li>• Proxy examines/processes requests for a website but a firewall does not (checks type of signal) // Proxy <b>processes</b> client-side requests whereas firewall <b>filters</b> packets</li> <li>• Proxy transmits website data to the user, but a firewall does not (it allows valid signals)</li> <li>• Proxy allows faster access to a web page using cache, but a firewall does not (allow faster access or have cache)</li> <li>• Proxy can hide internal network from internet, but a firewall cannot</li> </ul> | <b>3</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(d)(i)  | <p>Any <b>two</b> from:</p> <p>e.g.</p> <ul style="list-style-type: none"> <li>• Storing bookmarks/favourites</li> <li>• Storing history</li> <li>• Allow multiple tabs/web pages to be open</li> <li>• Allow movement back and forth between web pages // provides navigation tools</li> <li>• Allows the user to enter a URL/IP into the address bar</li> <li>• <b>Manages</b> HTTP/HTTPS protocol</li> <li>• Search cache for IP // Request IP from DNS // Send URL to DNS</li> <li>• Sends a request to the <b>IP address/web server</b> (to obtain the contents of a web page)</li> <li>• Runs active script/JavaScript/client-side script</li> <li>• Allows <b>files</b> to be downloaded from <b>website/internet</b></li> </ul> | <b>2</b> |
| 5(d)(ii) | <p>Any <b>two</b> from:</p> <p>e.g.</p> <ul style="list-style-type: none"> <li>• Storing preferences // so the user does not have to select their preferences each time they visit the site</li> <li>• Storing account details // so the user does not have to remember/enter their username and password each time they visit the site</li> <li>• Storing recent purchases // to allow the user to quickly re-order more items</li> <li>• Storing the pages visited/items selected // to display relevant adverts</li> <li>• Storing shopping basket // so when the user leaves the site the items are still in their basket</li> </ul>                                                                                                | <b>2</b> |

| Question | Answer                                                                                                                                                                                                                                                                                         | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 8(a)     | <b>Three</b> from: <ul style="list-style-type: none"> <li>• Trial and error to <b>guess a password</b></li> <li>• <b>Combinations</b> are repeatedly entered ...</li> <li>• ... until correct password is found</li> <li>• Can be carried out manually or automatically by software</li> </ul> | 3     |
| 8(b)(i)  | Any <b>two</b> from:<br>e.g. <ul style="list-style-type: none"> <li>• Steal/view/access data</li> <li>• Delete data</li> <li>• Change data</li> <li>• Lock account // Encrypt data</li> <li>• Damage reputation of a business</li> </ul>                                                       | 2     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 8(b)(ii) | Any <b>three</b> from:<br>e.g. <ul style="list-style-type: none"> <li>• Virus</li> <li>• Worm</li> <li>• Trojan horse</li> <li>• Spyware</li> <li>• Adware</li> <li>• Ransomware</li> </ul>                                                                                                                                                                                       | 3     |
| 8(c)     | Any <b>two</b> from: <ul style="list-style-type: none"> <li>• Two-step verification//Two-factor authentication//by example</li> <li>• Biometrics</li> <li>• Firewall // Proxy-server</li> <li>• <b>Strong/complex</b> password // by example</li> <li>• Setting a limit for login attempts</li> <li>• Drop-down boxes</li> <li>• Request for partial entry of password</li> </ul> | 2     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 10(a)    | <ul style="list-style-type: none"> <li>• To obtain <b>personal</b> data/details // by example</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| 10(b)    | <p><b>One</b> mark for each correct part of the diagram.<br/> Diagram shows:</p> <ul style="list-style-type: none"> <li>• User clicks/opens attachment/link that triggers download</li> <li>• Malicious software downloaded onto user's computer</li> <li>• User enters website address</li> <li>• User is <b>redirected</b> to fake website</li> </ul> <p>e.g.</p> <pre> graph TD     A[User clicks link that downloads malware to computer] --&gt; B[User computer malware]     B --&gt; C[User types in web address]     C -.-&gt; D[Real website]     C --&gt; E[Fake website]     E --&gt; F[Request gets redirected] </pre> | 4     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10(c)    | <b>Two</b> from:<br>Displays web pages<br>... by rendering HTML                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>2</b> |
| 10(d)    | Any <b>three</b> from:<br>e.g. <ul style="list-style-type: none"> <li>• Storing bookmarks/favourites</li> <li>• Recording user history</li> <li>• Allowing use of multiple tabs</li> <li>• Providing navigation tools // by example</li> <li>• Providing an address bar</li> <li>• Managing protocols // by example // checking digital certificate</li> <li>• Send URL to DNS</li> <li>• Sends a request to the <b>IP address/web server</b> (to obtain the contents of a webpage)</li> <li>• Runs active script/JavaScript/client-side script</li> <li>• Allows <b>files</b> to be downloaded from <b>website/internet</b></li> </ul> | <b>3</b> |
| 10(e)    | Any <b>four</b> from: <ul style="list-style-type: none"> <li>• Session cookies are stored in memory/RAM</li> <li>• ... whereas persistent cookies are stored on the hard drive/secondary storage</li> <li>• When the browser is closed a session cookie is lost</li> <li>• ... whereas a persistent cookie is not lost</li> <li>• ... until deleted by the user/they expire</li> </ul>                                                                                                                                                                                                                                                  | <b>4</b> |

## JUNE 2023 (2)

| Question | Answer                                                                                                                                                                                                                                | Marks    |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6(a)     | <b>One</b> mark for each correct term. <ul style="list-style-type: none"> <li>• Text</li> <li>• Web browser // web server</li> <li>• Web server // web browser</li> <li>• Session</li> <li>• Session</li> <li>• Persistent</li> </ul> | <b>6</b> |
| 6(b)     | Any <b>three</b> from:<br>e.g. <ul style="list-style-type: none"> <li>• Saving personal details</li> <li>• Storing login details</li> <li>• Tracking user preferences</li> <li>• Holding items in an online shopping cart</li> </ul>  | <b>3</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(a)     | <p><b>One</b> mark for each part of the diagram (MAX six).<br/> The diagram demonstrates:</p> <ul style="list-style-type: none"> <li>• Malware downloaded to several computers</li> <li>• ... turning it into a bot/zombie</li> <li>• ... creating a network of bots/zombies</li> <li>• Third party/hacker initiating the attack</li> <li>• <b>Bots</b> send requests to a web <b>server</b> at the same time</li> <li>• The web <b>server</b> fails due to the requests</li> <li>• Legitimate requests cannot reach the web server</li> </ul> | <b>6</b> |

| Question | Answer                                                                                                                                                                                                                                                | Marks    |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7(b)     | Any <b>two</b> from:<br>e.g. <ul style="list-style-type: none"> <li>• Revenge</li> <li>• To affect a company's reputation</li> <li>• Entertainment value</li> <li>• To demand a ransom to stop it</li> <li>• To test a system's resilience</li> </ul> | <b>2</b> |
| 7(c)     | Any <b>two</b> from: <ul style="list-style-type: none"> <li>• Proxy server</li> <li>• Firewall</li> <li>• Users scanning their computers with anti-malware</li> </ul>                                                                                 | <b>2</b> |

# JUNE 2023 (3)

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 6(a)     | Any <b>three</b> from:<br>e.g. <ul style="list-style-type: none"><li>• Checking the spelling and tone of the email/website</li><li>• Checking the URL attached to a link</li><li>• Scanning a download with anti-malware</li><li>• Only downloading data / software from trusted sources</li><li>• Never providing personal details online</li><li>• Install a firewall to check if the website is valid</li></ul> | 3     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 6(b)     | <b>Two</b> marks for description, <b>one</b> mark for example: <ul style="list-style-type: none"><li>• Manipulating / deceiving / tricking people ...</li><li>• ... to obtain data // to force them to make an error</li><li>• Any suitable example of social engineering</li></ul>                                                                        | 3     |
| 6(c)     | Any <b>three</b> from: <ul style="list-style-type: none"><li>• Providing users with different <b>permission</b> for the data</li><li>• Limiting access to reading data limiting the data that can be viewed</li><li>• Limiting access to editing data // limiting the data that can be deleted / changed</li><li>• Normally linked to a username</li></ul> | 3     |

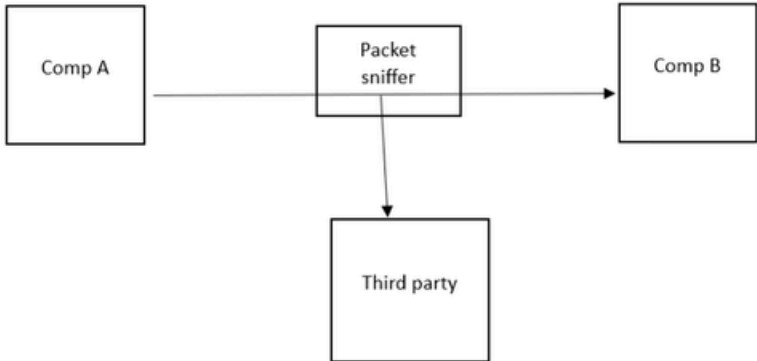
| Question | Answer                                                                                                                                                                                      | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 8        | <b>One</b> mark for each correct term in the correct order: <ul style="list-style-type: none"><li>• Malware</li><li>• Bot</li><li>• Botnet</li><li>• Web server</li><li>• Website</li></ul> | 5     |



| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 6        | <p>The diagram includes (any <b>four</b> from):</p> <ul style="list-style-type: none"> <li>– Traffic passing both ways through the firewall</li> <li>– An indication that criteria is set for the firewall</li> <li>– Traffic is compared to criteria</li> <li>– Traffic being rejected if it does/does not meet criteria</li> <li>– Traffic being accepted if it does/does not meet criteria</li> </ul> <p>e.g.</p> | 4     |

## NOVEMBER 2023 (2)

| Question | Answer                                                                       | Marks |
|----------|------------------------------------------------------------------------------|-------|
| 1(a)     | – A                                                                          | 1     |
| 1(b)     | Any <b>one</b> from:<br>– Spyware // Keylogger<br>– Adware<br>– Trojan horse | 1     |
| 1(c)     | – Anti-malware                                                               | 1     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 8(a)     | <p>The diagram demonstrates (<b>One</b> mark for each part of the diagram):</p> <ul style="list-style-type: none"> <li>– Data is being sent from one device to another</li> <li>– The data is being examined <b>during transmission</b></li> <li>– Packet sniffer is used</li> <li>– Intercepted data is reported to a third-party <b>during transmission</b> ...</li> <li>– ... and analysed for anything useful</li> <li>– Connection hacked to spoof destination address</li> </ul> <p>e.g.</p> <div style="text-align: center;"> <p>Packet sniffer examines data as it is sent from one computer to another</p>  <pre> graph LR     A[Comp A] --&gt; S[Packet sniffer]     S --&gt; B[Comp B]     S --&gt; T[Third party]             </pre> <p>Analyses data for anything useful</p> </div> | 4     |
| 8(b)     | <ul style="list-style-type: none"> <li>– Encryption ...</li> <li>– ... if the data is intercepted it will be <b>meaningless</b> (because they do not have the decryption key)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2     |

| Question                            | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      | Marks       |                |                                                     |        |                                                                                     |                                |                                                                                                                    |            |                                                                                          |                                   |                                                                           |                                     |                                                       |   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------|----------------|-----------------------------------------------------|--------|-------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------|---|
| 9                                   | <table><tr><th>Term</th><th>Description</th></tr><tr><td>world wide web</td><td>the collective name for all the web pages available</td></tr><tr><td>cookie</td><td>a small text file, stored by the web browser, that can store a user's personal data</td></tr><tr><td>uniform resource locator (URL)</td><td>the <b>text-based address</b> for a website // It is made up of the protocol, domain name and filename/folder name</td></tr><tr><td>web server</td><td>Stores <b>web pages</b> // receives requests from clients and returns requested web page</td></tr><tr><td>hypertext markup language // HTML</td><td>the language used to create a website. Example tags are &lt;head&gt; and &lt;body&gt;</td></tr><tr><td>hypertext transfer protocol // HTTP</td><td>a protocol that is used to request and send web pages</td></tr></table> | Term | Description | world wide web | the collective name for all the web pages available | cookie | a small text file, stored by the web browser, that can store a user's personal data | uniform resource locator (URL) | the <b>text-based address</b> for a website // It is made up of the protocol, domain name and filename/folder name | web server | Stores <b>web pages</b> // receives requests from clients and returns requested web page | hypertext markup language // HTML | the language used to create a website. Example tags are <head> and <body> | hypertext transfer protocol // HTTP | a protocol that is used to request and send web pages | 6 |
| Term                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |             |                |                                                     |        |                                                                                     |                                |                                                                                                                    |            |                                                                                          |                                   |                                                                           |                                     |                                                       |   |
| world wide web                      | the collective name for all the web pages available                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |             |                |                                                     |        |                                                                                     |                                |                                                                                                                    |            |                                                                                          |                                   |                                                                           |                                     |                                                       |   |
| cookie                              | a small text file, stored by the web browser, that can store a user's personal data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |             |                |                                                     |        |                                                                                     |                                |                                                                                                                    |            |                                                                                          |                                   |                                                                           |                                     |                                                       |   |
| uniform resource locator (URL)      | the <b>text-based address</b> for a website // It is made up of the protocol, domain name and filename/folder name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |             |                |                                                     |        |                                                                                     |                                |                                                                                                                    |            |                                                                                          |                                   |                                                                           |                                     |                                                       |   |
| web server                          | Stores <b>web pages</b> // receives requests from clients and returns requested web page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |             |                |                                                     |        |                                                                                     |                                |                                                                                                                    |            |                                                                                          |                                   |                                                                           |                                     |                                                       |   |
| hypertext markup language // HTML   | the language used to create a website. Example tags are <head> and <body>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |             |                |                                                     |        |                                                                                     |                                |                                                                                                                    |            |                                                                                          |                                   |                                                                           |                                     |                                                       |   |
| hypertext transfer protocol // HTTP | a protocol that is used to request and send web pages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |             |                |                                                     |        |                                                                                     |                                |                                                                                                                    |            |                                                                                          |                                   |                                                                           |                                     |                                                       |   |

| Question | Answer                                                                                                                                                                                                                  | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 12(a)    | Any <b>two</b> from: <ul style="list-style-type: none"> <li>– <u>Only</u> exists electronically</li> <li>– Can be a decentralised system</li> <li>– Can be a centralised system</li> <li>– Usually encrypted</li> </ul> | 2     |
| 12(b)    | – Blockchain                                                                                                                                                                                                            | 1     |

## NOVEMBER 2023 (3)

| Question | Answer                                                                                                                    | Marks    |
|----------|---------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a)     | Any <b>two</b> from: <ul style="list-style-type: none"><li>– Display web pages</li><li>– ... by rendering HTML</li></ul>  | <b>2</b> |
| 4(b)     | – Storing cookies                                                                                                         | <b>1</b> |
| 4(c)     | Any <b>one</b> from: <ul style="list-style-type: none"><li>– Storing bookmarks</li><li>– Storing favourites</li></ul>     | <b>1</b> |
| 4(d)     | Any <b>one</b> from: <ul style="list-style-type: none"><li>– It encrypts it</li><li>– Uses digital certificates</li></ul> | <b>1</b> |